# Youcef Ouadjer, Ph.D. Candidate

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## Research Profile :

- **Expertise:** Biometrics, pattern recognition, signal and image processing. Demonstrated advanced knowledge and innovative problem-solving abilities through rigorous research.

- **Research Design and Analysis**: Skilled in designing, executing, and analyzing complex experiments/studies using deep learning frameworks to generate impactful findings.

- **Scientific Communication**: Experienced in presenting complex concepts through compelling written reports, technical documents, and engaging oral presentations tailored to diverse audiences.

## Education:

- **National Polytechnic School** Algiers, Algeria

  *Ph.D. Student – Electrical and Electronics Engineering* *Nov. 2018 – Present*

- Thesis: Exploiting Behavioral Biometric Attributes for Active Authentication on Mobile Devices

  **Université Mouloud Mammeri Tizi Ouzou** Tizi Ouzou, Algeria

  *Master of science in Biomedical Engineering* *Sep. 2015- Jun. 2017*

- **Université Mouloud Mammeri Tizi Ouzou**

*Bachelor of Engineering in Electronics* *Sep. 2011- Jun. 2015*

## Research interests:

I am deeply passionate about advancing the field of behavioral biometrics and multimodal authentication systems, with a strong focus on developing secure, efficient, and user-friendly solutions for continuous authentication. My work bridges the gap between innovative machine learning methodologies and real-world applications, particularly in smartphone security. Leveraging techniques such as self-supervised learning, lightweight convolutional neural networks, and multimodal fusion frameworks, I aim to address pressing challenges in dataset synchronization, computational efficiency, and practical scalability. My research has culminated in the creation of novel datasets and robust models that highlight the potential of behavioral biometrics for next-generation authentication systems. I am committed to pushing the boundaries of artificial intelligence to solve critical problems at the intersection of security, usability, and technology.

## Future research directions:

- Adaptive authentication systems: Design adaptive systems that learn and evolve with user behavior over time, ensuring improved accuracy and robustness.
- Explainable and interpretable AI in biometrics: Explore methods to improve the interpretability of biometric authentication systems for increased trust and regulatory compliance.
- Multimodal Integration with Emerging Sensors: Expand research into combining traditional biometrics with new sensor modalities, such as wearable devices or environmental context sensing, for a more comprehensive authentication system.

## Academic and professional experience:

- **Ecole Nationale Polytechnique d'Alger** Algiers, Algeria
  *Graduate research assistant, Electrical Engineering* *Nov. 2018 - Present*
- **EURECOM Engineering School** Sophia Tech Campus, France
  Research internship. Jan 02 – Jan 31, 2023.
- **EURECOM Engineering School** Sophia Tech Campus, France
  Research internship Sep 30 – Oct 29, 2023

## Skills:

- **Programming:** Python, Java, C++ | **Frameworks:** Pytorch | **Code versioning:** Git
- **Languages:** Tamazight (*mother tongue*) Arabic (*fluent*), French (*fluent*), English (*fluent*).
- **App development:** Android studio.

## Publications:

- **Y. Ouadjer**, M. Adnane, N. Bouadjenek, Feature Importance Evaluation of Smartphone Touch Gestures for Biometric Authentication, in: 2nd International Workshop on Human-Centric Smart Environments for Health and Well-being, Boumerdes, Algeria, 2021.
- **Y. Ouadjer,** C. Galdi, S.-A. Berrani, M. Adnane, and J.-L. Dugelay, "Self-supervised based multimodal fusion for active biometric verification on mobile devices," in 2024 International Conference on Pattern Recognition Application and Methods, Rome, Italy, 2024.
- **Y. Ouadjer,** M.-A. Bellebna, S.-A. Berrani, M. Adnane, "Behavioral Biometrics: A multimodal Signal Processing Approach for User Authentication on mobile phones", International School on Signal Processing & its Applications, Algiers, June, 2024.
- **Y. Ouadjer**, S.-A. Berrani, N. Bouadjenek, M. Adnane, "User Authentication on Mobile Devices Based on Self-supervised Learning", International Journal of Biometrics, Accepted for publication, 2024.
- **Y. Ouadjer**, S.-A. Berrani, C. Galdi, M. Adnane, J.-L. Dugelay, ''MM-BioSync : A Multi-Modal Synchronized Sensing Dataset for Active Biometric Verification on Smartphones'', IET Biometrics (Under Review), 2025.
- H. Remma, C. Ouarezki, **Y. Ouadjer**, M. Adnane, S.-A. Berrani, ''Efficient Face Recognition Using Embedding-Based Distillation'', ENP Engineering Science Journal (Under Review), 2025.